Daily Sequences Drill #7 — Speed Sheet

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Mixed Synthesis Capstone · Every technique from Days 1–6, combined.

Section A Rapid Recognition

1. Write down the nth term of the AP $4, 9, 14, 19, \ldots$

Ans: $5n - 1$

2. Evaluate $\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5}$.

Ans: 32

3. A GP has $a = 6, r = \frac{1}{3}$. Find its sum to infinity.

Ans: 9

4. Write down the characteristic equation of $a_n = 7a_{n-1} - 10a_{n-2}$.

Ans: $x^2 - 7x + 10 = 0$

5. Find the fixed point of $f(x) = 3x - 8$.

Ans: $x = 4$

6. If the mean of the first 5 terms of a sequence is an integer, what can be said about their sum?

Ans: Their sum must be a multiple of 5.

7. Is $2, 6, 18, 54, \ldots$ an AP or a GP?

Ans: GP (ratio 3)

8. A recurrence has both characteristic roots with magnitude < 1. What happens to the sequence as $n \to \infty$?

Ans: Converges to 0.

9. Find the coefficient of x^2 in $(1+x)^5$.

Ans: 10

10. Write down the fixed-point formula for $f(x) = rx - c$ $(r \neq 1)$.

Ans: $x = \frac{c}{r-1}$

Section B Manipulation Drills

1. An AP has first term 2, common difference 3. Verify that it also satisfies the recurrence $a_n = 2a_{n-1} - a_{n-2}$.

2. Using $(1-x)^{-1} = \sum x^k$ and $x = \frac{1}{4}$, evaluate $\displaystyle\sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^k$.

A) $\frac{4}{3}$

B) $\frac{3}{4}$

C) 4

D) $\frac{1}{4}$

3. A recurrence has characteristic roots $\frac{1}{2}$ and $\frac{1}{3}$, with no forcing term. Does the sequence converge, and to what?

A) Converges to 0.

B) Converges to $\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$.

C) Diverges.

D) Oscillates without converging.

Ans: A) Converges to 0.

4. A sequence satisfies $a_{n+1} = \dfrac{1}{2 - a_n}$. Find its fixed point(s), and determine whether $a_1 = 1$ leads to a constant sequence.

Ans: $x = 1$

5. The sum of a binomial row, $\sum_k \binom{n}{k}$, and the sum of an AP's terms, $\sum_k a_k$, are both examples of what general mathematical operation?

A) Summing a finite sequence.

B) They are unrelated concepts.

C) Both are geometric series.

D) Both require calculus.

Ans: A) Summing a finite sequence.

6. In the alternating multiply-subtract process (Day 6), the shift $b_n = a_n - \frac{c}{r-1}$ turns Alice's map into exactly which type of sequence from Day 3?

A) A GP with ratio r.

B) An AP with difference r.

C) A GP with ratio c.

D) Neither.

Ans: A) A GP with ratio r.

7. The series $(1-x)^{-1} = \sum x^k$ is the closed form of which simple recurrence? State the recurrence and its characteristic root.

Ans: $a_k = xa_{k-1}$

8. Which of Day 1's arithmetic sequences is also strictly monotonic (Day 5's sense) for every valid common difference $d \neq 0$?

A) Every AP with $d \neq 0$ is monotonic.

B) Only APs with $d > 0$.

C) No AP is ever monotonic.

D) Only APs with integer d.

Ans: A) Every AP with $d \neq 0$ is monotonic.

9. A GP has $|r| > 1$. Is the sequence bounded?

A) No — it diverges in magnitude, unbounded.

B) Yes, always bounded.

C) Bounded only if $a > 0$.

D) Bounded only if r is an integer.

Ans: A) No, unbounded.

10. Day 6's "$a_k = 1$ for all k" construction and Day 2's binomial row-sum fact $\sum_k \binom{n}{k} = 2^n$ are both about sums. Are these two facts related, or independent observations?

Ans: Independent observations.

Section C Substitution & Structure

1. An AP and a GP both have first term 3; the AP has $d = 4$, the GP has $r = 2$. After how many terms (if ever) could they next coincide?

A) They never coincide again.

B) At term 4.

C) At term 6.

D) At term 10.

Ans: A) They never coincide again.

2. Compare the growth of $\sum_{k=1}^{n} a_k$ where $a_k = 1$ for all k (Day 6's construction) against $\sum_{k=0}^{n} \binom{n}{k}$ (Day 2's row sum) as $n \to \infty$. Which grows faster?

A) The binomial row sum grows faster (exponentially vs. linearly).

B) They grow at the same rate.

C) The Day 6 sum grows faster.

D) Neither grows — both are constant.

Ans: A) The binomial row sum grows faster.

3. A recurrence $a_n = pa_{n-1} + qa_{n-2}$ has both characteristic roots equal to a Day 6-style fixed point $\frac{c}{r-1}$. If $\frac{c}{r-1} = 1$, what does Day 4's result about repeated roots equal to 1 tell us about the long-term behaviour of (a_n)?

A) Linear growth in n (effectively an AP).

B) Exponential growth.

C) Convergence to 0.

D) Periodic oscillation.

Ans: A) Linear growth in n.

4. Evaluate $\displaystyle\sum_{k=0}^{\infty}(k+1)\left(\frac{1}{3}\right)^k$ using Day 2's identity $(1-x)^{-2} = \sum_{k=0}^{\infty}(k+1)x^k$.

A) $\frac{9}{4}$

B) $\frac{3}{2}$

C) $\frac{4}{9}$

D) $\frac{2}{3}$

Ans: A) $\frac{9}{4}$

5. A Möbius map (Day 5) is used as the combined operation in an alternating writing process (Day 6). If the map has order exactly 3 under iteration, what is guaranteed about EVERY starting value (where the map is defined)?

A) Every starting value returns to itself after exactly 3 applications.

B) Only some starting values return; others diverge.

C) The map converges to a fixed point instead.

D) Nothing general can be said.

Ans: A) Every starting value returns to itself after exactly 3 applications.

6. For a GP with ratio r, for which values of r is the sequence BOTH monotonic AND convergent?

A) $0 \leq r \leq 1$.

B) $-1 < r < 1$.

C) Only $r = 1$.

D) $r > 1$ only.

Ans: A) $0 \leq r \leq 1$.

7. Day 6's two-branch search ($u(r^k - 1) = -c$) and Day 4's root-magnitude convergence condition are both examples of what general principle about a term r^k (or r^n)?

A) Whether $|r| < 1$, $= 1$, or > 1 entirely determines the long-term size of r^k — shrinking to 0, staying constant, or blowing up.

B) Exponential terms are always eventually periodic.

C) Exponential terms never affect convergence.

D) There is no shared principle.

Ans: A) $|r| < 1, = 1, > 1$ entirely determines the size of r^k.

8. Which set of results from this week are all secretly governed by the SAME underlying idea — long-term behaviour controlled by whether a key magnitude is < 1, $= 1$, or > 1?

A) Geometric series convergence (Day 3); recurrence convergence via characteristic roots (Day 4); GP boundedness (today's B9).

B) Arithmetic sequences, binomial coefficients, integer means.

C) All seven days equally.

D) None of the sheet's content shares a common idea.

Ans: A) Geometric series convergence, recurrence convergence, GP boundedness.

Section D Challenge

1. For each positive integer $n \geq 4$, define an n-necklace to be a circular arrangement of n (not necessarily distinct) positive integers such that the product of every 4 neighbouring integers equals n. Determine the number of integers n in the range $4 \leq n \leq 1000$ for which it is possible to form an n-necklace. *(after BMO1 2019 Q2)*

Ans: 252

2. Prove that for any starting value a_1 (a positive integer), there exists a strictly increasing sequence of positive integers $a_1 < a_2 < \cdots < a_n$, of any desired length n, such that every partial sum S_k is divisible by k.

Ans: Proof: see method.

3. In Day 6's alternating recurrence, suppose the subtracted term is not a fixed constant c but grows every turn, $c_n = cn$. Does the shift-to-fixed-point trick (a single constant $L = \frac{c}{r-1}$) still work directly?

A) No — a single fixed constant L cannot absorb a forcing term that changes every step; a particular-solution technique (Day 4's C5) matching the forcing term's form would be needed instead.

B) Yes, exactly the same L still works.

C) The process becomes undefined.

D) It only works if $r = 1$.

Ans: A) No — a particular-solution technique is needed instead.

4. A sequence is defined by $a_1 = 1$ and, for $n \geq 1$, a_{n+1} is the SMALLEST positive integer greater than a_n such that S_{n+1} is divisible by $n + 1$. Find $a_1, \ldots, a_5$, and conjecture (with justification) a general closed form for a_n.

Ans: $a_n = 2n - 1$

5. Looking back across all seven days: which day's toolkit would be LEAST directly useful for tackling D1's n-necklace problem?

A) Day 2 (Binomial & Trinomial Expansion) — D1's argument uses periodicity and perfect-power number theory, not binomial coefficients.

B) Day 5 (Behaviour of Sequences).

C) Day 1 (Arithmetic Sequences).

D) All seven days are equally essential to D1.

Ans: A) Day 2 (Binomial & Trinomial Expansion).